Investment Management — Theory Compendium

Every concept you may be asked to explain, define or discuss · Q4-style written questions

Contents

PART 1 — Investment Fundamentals **2**

PART 2 — Debt Markets and Instruments **4**

PART 3 — Bond Analytics: the Concepts (maths in Guide Part 3–4) **6**

PART 4 — Market Efficiency (deep theory) **8**

PART 5 — Behavioural Finance (deep theory) **10**

PART 6 — Market Microstructure and Trading **11**

PART 7 — Rapid-fire concept bank (for Q4 "short notes") **13**

What this is. The guide teaches you *how to answer* and the practice set lets you *test the maths*. This document is the **concept bank**: crisp, exam-ready explanations of every theoretical topic on your syllabus — especially the definitional and "discuss/explain" material that pure formulas cannot cover. For a written answer, give a **one-line definition, two or three key points, and one illustration** — that structure earns full marks. Cross-references to the main guide are noted where the maths lives there.

PART 1 — Investment Fundamentals

1.1 Investment vs speculation

An **investment** commits money for an expected return commensurate with its risk, over a reasonable horizon, based on analysis. **Speculation** seeks large gains from short-term price moves, accepting high risk and relying on timing rather than fundamentals. The line is one of *degree* — risk taken, horizon, and basis for the decision.

1.2 Asset classes

An **asset class** is a group of investments with similar characteristics, behaviour and regulation. Classifying assets matters because each class has a distinct **risk-return profile** and they do not move together, so combining them **diversifies** risk. The main classes:

- **Equities (stocks):** ownership; highest expected return and risk; returns from capital gains + dividends.
- **Fixed income (bonds):** lending; lower risk, predictable income; sensitive to interest rates and credit.
- **Cash / money-market:** T-bills, CDs; very low risk and return; provide liquidity and safety.
- **Real estate:** income + appreciation; illiquid; inflation hedge.
- **Commodities:** gold, oil; inflation hedge; no income, price-driven.
- **Alternatives:** private equity, hedge funds, derivatives; low correlation, high complexity/illiquidity.

Key point for answers: the value of asset-class thinking is **diversification** — because classes have **low correlation**, mixing them lowers portfolio risk for a given return.

1.3 Return concepts (recap — maths in Guide Part 2)

Distinguish: **expected (arithmetic mean)** vs **realised (geometric mean/HPR)**; **gross vs net** (net = gross − transaction − information costs); **nominal vs real** (Fisher: real ≈ nominal − inflation); **pre- vs post-tax**. Always state which one you are using.

1.4 Types of risk — systematic vs unsystematic (very common theory question)

Total risk splits into two parts:

- **Systematic (market / non-diversifiable) risk:** affects the whole market — interest rates, inflation, recession,policy changes, war. It **cannot be diversified away** and is measured by **beta**. Investors are *compensated* only for this risk.
- **Unsystematic (specific / diversifiable) risk:** unique to a firm or industry — a strike, a lawsuit, a product failure. It **can be eliminated by diversification**, so the market does *not* reward bearing it.

$$\text{Total risk} = \text{Systematic risk} + \text{Unsystematic risk}$$

As you add stocks to a portfolio, unsystematic risk falls toward zero and only systematic (market) risk remains. This is *the* reason diversification works and why beta (systematic risk) — not total SD — drives expected return in the CAPM: $E(R) = R_f + \beta\,(R_m - R_f)$.

Sources of risk to be able to list: interest-rate, market, inflation/purchasing-power, business, financial (leverage), liquidity, exchange-rate, and country/political risk.

1.5 Risk aversion and utility (theory the "Risk Aversion" reading expects)

A **risk-averse** investor, offered two choices with the same expected return, prefers the one with **lower risk**; equivalently, she requires **extra expected return (a risk premium)** to take on more risk. Most investors are risk-averse.

- **Risk-averse:** dislikes risk; needs compensation for it (concave utility function — diminishing marginal utility of wealth).
- **Risk-neutral:** cares only about expected return, indifferent to risk.
- **Risk-seeking:** will accept a lower expected return for a chance at a big gain (e.g. lottery buyers).

Related ideas:

- **Risk premium:** the extra expected return demanded above the risk-free rate for bearing risk.
- **Certainty equivalent:** the guaranteed amount an investor would accept in place of a risky gamble; for a risk-averse person it is *less* than the gamble's expected value, and the gap is the risk premium.
- **Indifference curves:** in risk-return space, a risk-averse investor's indifference curves slope **upward** (more risk must be paid for with more return) and steeper curves mean greater aversion. The optimal portfolio is where the highest indifference curve touches the efficient frontier.

How it's asked: "What does it mean that an investor is risk-averse? How does it affect required return?" → define, mention the risk premium and concave utility, illustrate with a certainty-equivalent example.

PART 2 — Debt Markets and Instruments

2.1 Money-market instruments (short-term, < 1 year, very liquid)

- **Treasury Bills (T-bills):** government, < 1 year, issued at a discount, **sovereign (risk-free)**, highly liquid.
- **Commercial Paper (CP):** unsecured short-term corporate borrowing for working capital; only strong-credit firms.
- **Certificate of Deposit (CD):** negotiable bank deposit; a bank liability used for funding.
- **Repo (repurchase agreement):** sell a security now and buy it back later at a set price — effectively a **collateralised short-term loan**; used by banks/RBI to manage liquidity.
- **Call money:** very short (overnight) interbank lending.

2.2 Capital-market debt (long-term)

- **Government Securities (G-secs):** long-dated sovereign bonds; benchmark for the risk-free curve.
- **State Development Loans (SDLs):** state-government bonds; quasi-sovereign.
- **Corporate bonds / NCDs (non-convertible debentures):** company borrowing for expansion; carry credit risk and pay a spread over G-secs.
- **External Commercial Borrowings (ECB):** foreign-currency borrowing from foreign lenders.

2.3 Bond terminology (recap)

Face/par value (repaid at maturity), **coupon rate** (annual interest as % of face), **maturity date**, and the **indenture** (the legal contract between issuer and holder).

2.4 Types of bonds

- **By coupon:** *fixed* (constant coupon); *floater* (coupon tied to a benchmark like the repo rate — rises/falls with it); *inverse floater* (coupon moves *opposite* the benchmark — double hit when rates rise); *zero-coupon* (no coupons, sold at a discount).
- **By redemption feature:** *callable* (issuer may repurchase early — bad for the holder, so it pays a **higher** coupon; issuers call when rates fall); *puttable* (holder may sell back early — good for holder, so **lower** coupon); *convertible* (holder may swap for a set number of shares).
- **International:** *Eurobond* (issued outside the country of its currency, e.g. a USD bond sold in Europe); *foreign bond* (issued in a domestic market by a foreigner, in that market's currency) — nicknamed **Yankee** (USD in US), **Samurai** (JPY in Japan), **Bulldog** (GBP in UK), **Kangaroo** (AUD), **Masala** (INR bond sold abroad, shifting FX risk to the investor).

2.5 Bond innovations (know one line each)

- **TIPS / index bonds:** principal and/or coupon linked to inflation (CPI) — protect real value.

- **Catastrophe bonds:** high coupon, but principal is forfeited if a defined disaster occurs — transfers insurance risk to bondholders.
- **Pay-in-kind (PIK):** issuer may pay interest in *more bonds* instead of cash during a cash crunch.
- **Asset-backed securities (ABS/MBS):** coupons serviced by cash flows from a pool of assets (mortgages, loans).
- **Sovereign Gold Bonds (SGB):** RBI, 2.5% coupon plus redemption linked to the gold price — a paper substitute for physical gold.

2.6 Indentures and protective covenants

The **indenture** protects bondholders through clauses such as:

- **Sinking fund:** issuer must retire part of the debt periodically before maturity — reduces default risk.
- **Subordination:** limits additional borrowing and ranks senior bondholders ahead of juniors in bankruptcy.
- **Collateral:** specific assets pledged; a **secured** bond has collateral, a **debenture** is unsecured (backed only by the issuer's general creditworthiness).
- **Protective (restrictive) covenants:** limits on dividends, further debt, asset sales.

2.7 Bond safety, ratings and credit spread

Analysts judge safety with financial ratios: **coverage** (earnings ÷ fixed charges), **leverage** (debt ÷ equity), **liquidity** (current, quick ratios), **profitability**, and **cash-flow-to-debt**. **Rating agencies** (S&P, Moody's, CRISIL, ICRA) grade this:

- **Investment grade:** BBB/Baa and above.
- **Speculative / "junk":** BB/Ba and below.

The **credit spread** = corporate yield − government yield = compensation for **default risk**; it widens as the estimated default probability rises. *Watch-out term:* **ratings shopping** — an issuer choosing the agency likely to give the best rating.

2.8 Risks in bonds (be able to list and one-line each)

Interest-rate risk (price falls when rates rise), **reinvestment risk** (coupons reinvested at lower rates), **credit/default risk**, **inflation (purchasing-power) risk**, **call risk**, and **liquidity risk**. *Note the see-saw:* interest-rate risk and reinvestment risk work in **opposite** directions — the basis of immunisation (Guide Part 4).

PART 3 — Bond Analytics: the Concepts (maths in Guide Part 3–4)

3.1 The price–yield relationship and its rules

Price and yield move **inversely** (yield is the discount rate). The key qualitative rules (useful as a written answer):

1. Price moves inversely to yield.
2. **Longer maturity → greater price sensitivity.**
3. **Lower coupon → greater price sensitivity.**
4. Sensitivity increases with maturity but at a **decreasing rate**.
5. A yield *fall* raises price **more** than an equal yield *rise* lowers it (**convexity/asymmetry**).

3.2 Duration — meaning and uses

Duration is the **weighted-average time to receive a bond's cash flows** (weights = PV of each flow) — think of it as the *balance point* of the cash flows, and as the bond's **effective maturity**. Its main use is as the measure of **interest-rate sensitivity** (via modified duration) and as the tool for **immunisation**. Properties: rises with maturity, falls with higher coupon and higher yield; a zero's duration = its maturity; a perpetuity's = $(1 + y)/y$.

3.3 Convexity — meaning

Convexity measures the **curvature** of the price-yield relationship (how duration itself changes as yields change). Because duration is only a straight-line estimate, convexity corrects it for large moves. **Positive convexity is desirable:** the bond falls *less* than duration predicts when yields rise and gains *more* when they fall.

3.4 Immunisation — concept

A strategy that makes a bond portfolio's value **insensitive to interest-rate changes** by setting the **asset duration equal to the liability's horizon** (and matching present values). It works because a rate change's effect on **price** and on **reinvestment** income then cancel out. It is **dynamic** — durations drift, so the portfolio must be **rebalanced**.

3.5 Term-structure theories (why the yield curve has its shape)

- **Pure Expectations:** long rates = the geometric average of *expected future short rates*; the curve's slope reflects where the market thinks short rates are heading.
- **Liquidity Preference:** expectations **plus a liquidity premium** rising with maturity (long bonds carry more price risk); implies an upward-sloping curve **even if** rates are expected to stay flat.
- **Segmented Markets / Preferred Habitat:** supply and demand within each maturity "bucket" set its rate; investors/issuers prefer maturities matching their liabilities but can be tempted out for enough extra yield.

3.6 Yield-curve shapes and the economy

Upward (normal): expansion / expected higher rates. **Inverted** (long < short): expected rate cuts, a classic **recession signal** (typically ~4–5 quarters ahead). **Flat:** transition. The

term spread (e.g. 10-year minus 3-month) is used to forecast growth.

3.7 Clean vs dirty price (recap)

Dirty (what you pay) = Clean (what is quoted) + Accrued interest. Clean prices allow fair comparison across time; dirty is the true cash cost. *(Full example in Guide 5.3.)*

3.8 Active vs passive bond management

- **Passive:** *buy-and-hold* (match desired credit/duration and hold) or *indexing* (track a bond index). Assumes efficiency.
- **Active:** *interest-rate anticipation* (lengthen duration if you expect rates to fall, shorten if you expect a rise) and *credit analysis* (act before the rating agencies).

PART 4 — Market Efficiency (deep theory)

4.1 Definition and the underlying model

An **efficient market** is one in which security prices **fully and instantly reflect all available information**, so a price is an **unbiased estimate of intrinsic value**. Formally it is a **"fair game"**: you cannot systematically earn returns above those justified by risk. A corollary is the **random walk** — because prices move only on *new* (unpredictable) information, successive price changes are essentially independent.

4.2 Fama's necessary conditions for efficiency

Efficiency is *sufficient* (though not strictly necessary) if: (i) **no transaction costs**, (ii) **information is freely available to all**, and (iii) **all investors agree on the implications** of information for prices. Real markets violate these to a degree — which is exactly why *some* inefficiency can persist (see 4.7).

4.3 The three forms (Fama's taxonomy)

Form	Information already in the price	Implication
Weak	All past prices and trading volumes	**Technical analysis** cannot earn excess returns
Semi-strong	All *publicly available* information	**Fundamental analysis** cannot earn excess returns either
Strong	*All* information, public and **private/insider**	Not even insiders can profit

Each form nests the previous one. Consensus: developed markets are weak- and largely semi-strong-efficient, but **not strong-form** (insiders demonstrably profit, which is why insider trading is illegal).

4.4 Implications of efficiency

- **For investors:** hard to beat the market → favour **low-cost indexing**, diversification, and minimal trading; timing and stock-picking rarely pay after costs.
- **For analysts:** research only adds value where it produces genuinely new insight; most public analysis is already priced in.
- **For companies:** the market fairly values securities, so financial "gimmicks" (e.g. cosmetic accounting) do not fool it; the timing of issues cannot reliably exploit mispricing.

What efficiency does NOT mean: that prices never change, that no one ever profits, or that price = value with zero error at every instant. It means **no *systematic, exploitable* mispricing after costs**.

4.5 Tests of efficiency

- **Weak-form:** *serial-correlation/autocorrelation* tests (are returns predictable from past returns?), *runs tests* (are up/down streaks random?), *filter-rule* tests (do

mechanical buy/sell rules beat buy-and-hold?). *(Maths in Practice P14.)*
- **Semi-strong:** *event studies* (does the price adjust instantly to an announcement?) and *portfolio/anomaly studies* (do stocks sorted on P/E, size, etc. earn abnormal returns?). *(Event study concept in Guide 5.4; maths in Practice P15.)*
- **Strong-form:** *insider-trading* studies and *mutual-fund performance* studies (do professionals beat the market?).

4.6 The joint-hypothesis problem (a subtle, high-value point)

Any test of efficiency is really a **joint test** of (i) market efficiency **and** (ii) the asset-pricing model used to define "normal"/expected returns. If a test finds "abnormal" returns, you cannot be sure whether the market is *inefficient* or whether your **model of expected return was simply wrong**. This is why efficiency is so hard to prove or refute definitively.

4.7 Why pockets of inefficiency persist

- **Grossman-Stiglitz paradox:** if prices already reflected *all* information, no one would be paid to gather it, and prices would stop being efficient. So a little exploitable mispricing **must** remain to reward the analysts who keep markets efficient. A gap smaller than trading costs is not a true inefficiency.
- **Self-correcting mechanism:** when a profitable pattern is found, investors trade on it and *arbitrage it away* — so anomalies tend to shrink over time (illustration: **Numeric Investors** exploiting analyst underreaction until competition cut the window from weeks to minutes).

4.8 Documented anomalies (evidence of inefficiency — be able to name a few)

- **Calendar effects:** the **January effect** and the **weekend/Monday effect**.
- **Size effect:** small-cap stocks earn higher risk-adjusted returns.
- **Value effect:** low **P/E** and low price-to-book stocks outperform.
- **Momentum:** past winners keep winning over 3–12 months.
- **Post-earnings-announcement drift (PEAD):** prices keep drifting in the direction of an earnings surprise for weeks — a direct semi-strong violation.
- **Overreaction (De Bondt–Thaler):** past extreme losers later outperform past winners (long-run reversal).
- **Neglected-firm effect:** under-researched stocks earn excess returns.

PART 5 — Behavioural Finance (deep theory)

5.1 Why it exists

Classical finance assumes rational, utility-maximising investors. Behavioural finance shows people are **boundedly rational** and make **systematic** (predictable) errors, and that **limits to arbitrage** stop smart money from fully correcting the resulting mispricings. Together these produce the anomalies and "pockets of inefficiency."

5.2 Prospect theory (Kahneman & Tversky) — the core model

Describes how people *actually* decide under risk, versus rational expected-utility theory:

- **Reference dependence:** people judge outcomes as **gains/losses from a reference point** (often the purchase price), not final wealth.
- **Loss aversion:** losses hurt about **twice** as much as equal gains please — the value function is steeper for losses.
- **Diminishing sensitivity:** the value function is concave for gains, convex for losses → people are **risk-averse in gains but risk-seeking in losses** (they gamble to avoid realising a loss).
- **Probability weighting:** people **overweight small probabilities** (buy lotteries and insurance) and underweight large ones.

5.3 Key biases — explained, not just listed

- **Overconfidence:** overrating one's own information/skill → **overtrading** and under-diversification.
- **Anchoring:** over-relying on a reference number (e.g. the price paid) when updating beliefs.
- **Representativeness:** judging by stereotypes/recent patterns → chasing "hot" stocks, ignoring base rates.
- **Availability:** overweighting easily-recalled, vivid events (e.g. fearing a recent-crash scenario).
- **Herding:** following the crowd → **bubbles and crashes**.
- **Mental accounting:** treating money differently by arbitrary "account" (e.g. spending a bonus recklessly).
- **Disposition effect:** selling **winners too early** and holding **losers too long** (a consequence of loss aversion).
- **Framing:** the *way* a choice is presented changes the decision.
- **Conservatism / under-reaction:** updating beliefs too slowly on new information → PEAD, momentum.
- **Gambler's fallacy & hindsight bias:** expecting reversals in random sequences; believing, after the fact, that outcomes were predictable.

5.4 Limits to arbitrage

Even when rational traders spot mispricing, they may not correct it because of: **fundamental risk** (the misvaluation can worsen before it corrects), **noise-trader risk** (irrational traders push prices further away — "markets can stay irrational longer than you can stay solvent"), **implementation costs** and **short-selling constraints**. This is why behavioural mispricings can persist — the link back to *pockets of inefficiency*.

PART 6 — Market Microstructure and Trading

6.1 Types of markets

- **Primary vs secondary:** primary = new issues (IPOs, new bond sales) raise capital for the issuer; secondary = trading of existing securities among investors (provides **liquidity** and price discovery).
- **Order-driven (auction) vs quote-driven (dealer):** in an order-driven market buyers and sellers are matched on a central order book (most stock exchanges); in a quote-driven market **dealers/market-makers** post bid and ask prices and trade from their own inventory (common in bonds/FX).

6.2 Order types

- **Market order:** execute **immediately** at the best available price — prioritises speed, gives no price control (you pay the spread/impact).
- **Limit order:** execute only at your specified price **or better** — price control, but may not fill.
- **Stop-loss order:** dormant until a trigger price is hit, then becomes a **market order** — used to cap losses.
- **Stop-limit / trailing stop:** a stop that converts to a limit order, or one whose trigger follows the price to lock in gains.
- **Duration qualifiers:** *day* orders (expire end of day) vs *good-till-cancelled (GTC)*; *all-or-none (AON)*, *fill-or-kill (FOK)*.

6.3 Bid-ask spread and its determinants

The **spread** = ask − bid is the price of **immediacy** — the cost of trading right now against a market-maker. It has three components (know these — a likely theory question):

1. **Order-processing costs:** the dealer's overheads of executing trades.
2. **Inventory (holding) costs:** the dealer's risk of holding the security while waiting for the other side.
3. **Adverse-selection (information) costs:** the risk of trading against a **better-informed** counterparty; the dealer widens the spread to cover being picked off by insiders.

A **narrow spread signals a liquid, actively-traded, information-symmetric security;** wide spreads signal illiquidity or high information risk.

6.4 Impact cost and liquidity

A large order cannot all fill at the best price — it **"walks the book,"** filling at progressively worse prices. **Impact cost** = (average execution price − mid-price) ÷ mid-price measures this slippage. It is the practical face of **liquidity**: liquid securities have low spreads *and* low impact cost. *(Maths in Practice P16.)*

6.5 Transaction costs

- **Explicit (visible):** brokerage/commissions, exchange fees, **taxes (STT)**, and the bid-ask spread.

- **Implicit (hidden): market impact**, **slippage**, and **opportunity/delay cost** (price moving while you wait to trade).

Transaction costs **reduce net returns** and are central to the **active-vs-passive** debate: frequent active trading racks up costs that a low-turnover index fund avoids. Reduce them by trading less, using limit orders, and splitting large orders.

6.6 Short selling — mechanism, uses, risks

- **Mechanism: borrow** the shares, **sell** them now, later **buy them back** ("cover") and return them — profiting if the price **falls**.
- **Uses:** *speculation* (betting on a fall), *hedging* (offsetting a long position), and *arbitrage* (exploiting relative mispricing).
- **Risks:** the loss is **UNLIMITED** (the price can rise without bound) while the gain is capped at 100%; plus **margin calls**, a **short squeeze** (a rising price forcing shorts to buy, pushing it higher still), forced **buy-in** if the lender recalls shares, and paying any **dividends** to the lender.
- **Role in efficiency:** short-sellers aid **price discovery** by pushing overpriced stocks down; **constraints on short-selling** are a recognised *limit to arbitrage* that lets overpricing persist.

6.7 Margin trading (brief)

Buying with borrowed money amplifies both gains and losses; a fall in value triggers a **margin call**. Like shorting, it magnifies risk.

PART 7 — Rapid-fire concept bank (for Q4 “short notes”)

One-line skeletons you can expand with an illustration. Aim for definition → mechanism → example → significance.

- **Why price and yield move inversely:** yield is the discount rate; a higher rate lowers the PV of fixed future cash flows, so price falls.
- **Modified duration vs maturity:** maturity ignores interim coupons; duration weights all cash flows by PV and converts to price sensitivity — so it correctly ranks interest-rate risk.
- **Convexity’s value:** it corrects duration’s straight-line error and is always positive for plain bonds (cushions falls, boosts rises).
- **Immunisation:** match asset duration to the liability horizon so price and reinvestment risks cancel; rebalance over time.
- **Why the yield curve usually slopes up:** liquidity premium — investors demand extra yield for the greater price risk of long bonds, even absent expected rate rises.
- **Inverted curve:** market expects rate cuts / recession; a leading indicator.
- **Weak vs semi-strong vs strong efficiency:** past prices / all public info / all info incl. insider — killing technical, fundamental, and even insider advantage respectively.
- **Random walk:** prices move only on new (unpredictable) information, so changes are independent.
- **Joint-hypothesis problem:** an efficiency test is also a test of the assumed return model, so “abnormal returns” are ambiguous.
- **Pockets of inefficiency:** Grossman-Stiglitz — some mispricing must persist to reward information gathering.
- **Event study:** measures abnormal return (actual − market-model expected) around news; instant jump then flat = semi-strong efficient; drift = inefficiency; pre-move = leakage.
- **Clean vs dirty price:** dirty = clean + accrued interest; buyer pays dirty, market quotes clean.
- **Systematic vs unsystematic risk:** market-wide (undiversifiable, priced, = beta) vs firm-specific (diversifiable, not priced).
- **Risk aversion:** demands a risk premium for extra risk; certainty equivalent < expected value of a gamble.
- **Bid-ask spread determinants:** order-processing, inventory, and adverse-selection (information) costs.
- **Short selling:** borrow-sell-buyback; unlimited loss, capped gain; aids price discovery.
- **Passive vs active:** if markets are efficient, low-cost indexing beats the average active manager after costs; active pays only with a genuine edge.
- **Behavioural biases → anomalies:** loss aversion, overconfidence, herding and underreaction create momentum, bubbles and PEAD that limits-to-arbitrage let persist.

*You now have three complementary documents: the **Guide** (understand + how to answer), the **Practice Set** (test the maths), and this **Theory Compendium** (every concept for the written questions). Between them, the syllabus is covered end to end. In the exam, define crisply, give one illustration, and always show your interpretation and approach — that is what earns the marks. All the best.*